Evidence In Medicine

Thinking Critically About Evidence

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AY 2025-2026

Learning Objectives

After this session, you will be able to:

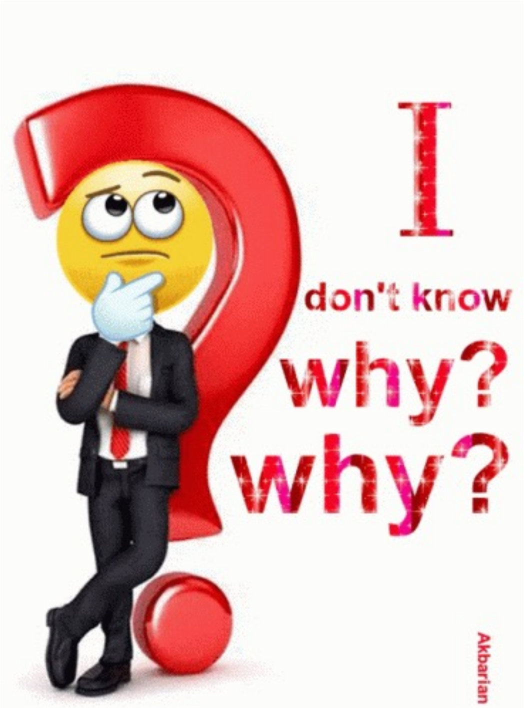
1. Explain what “evidence” means in medicine.
2. Identify different types and strengths of medical evidence.
3. Understand the meaning of Evidence-Based Medicine (EBM).
4. Explain what clinical reasoning means.
5. Recognize how critical thinking helps doctors use evidence.

What Is Evidence?

- **Evidence** means *facts or information* that help us decide if something is true or false.
  In medicine, evidence helps us know whether a **treatment, test, or theory really works.**
- 🩺 *Example:*
  If many studies show that walking 30 minutes daily lowers blood pressure, that is medical evidence.

Why Evidence Matters

- Doctors must make **safe, correct, and fair** decisions.
- Evidence helps avoid **guesswork or old habits.**
- It protects patients from **harmful or useless treatments.**
- Strong evidence = better medical outcomes.



Sources of Medical Evidence

Evidence in medicine can come from:

1. **Research studies** – scientific experiments or trials.
2. **Clinical experience** – what doctors observe in real patients.
3. **Patient data** – reports, symptoms, test results.
4. **Laboratory research** – cell or animal studies.
5. **Expert opinions** – advice from experienced doctors.

Types of Evidence

Evidence can be described as:

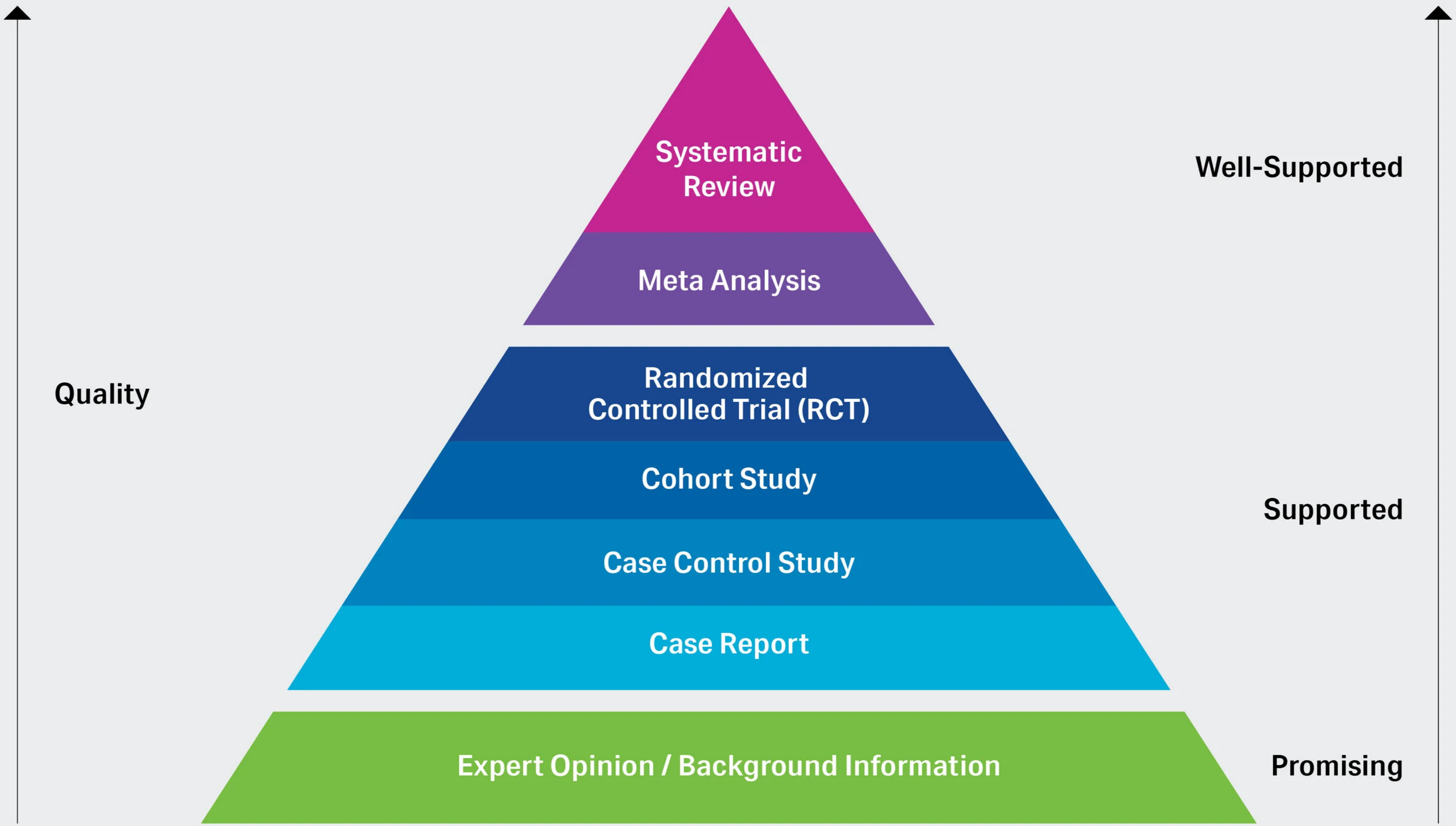
- **Quantitative:** based on numbers or measurements (e.g. blood sugar).
- **Qualitative:** based on feelings or opinions (e.g. how a patient feels).
- **Primary:** original study or data (e.g. one experiment).
- **Secondary:** summary or review of many studies.

The Hierarchy of Evidence

Some evidence is stronger than others.
From **strongest to weakest:**

1. Systematic reviews & meta-analyses
2. Randomized controlled trials (RCTs)
3. Cohort studies
4. Case-control studies
5. Case reports or case series
6. Expert opinion

Hierarchy of Evidence



Level 1 – Systematic Review / Meta-Analysis

- **Definition:**
  A systematic review collects and analyzes results from many studies on the same topic.
  A meta-analysis combines their data using statistics.
- **Example:**
  A meta-analysis of 20 studies shows that statins lower heart-attack risk.
- **Strength:**
  Highest level — gives the most reliable answer.

Level 2 – Randomized Controlled Trial (RCT)

- **Definition:**
  An RCT compares two or more groups of patients randomly. One group gets the new treatment; another gets a placebo or standard care.
- **Example:**
  Testing a new blood-pressure pill vs. placebo.
- **Strength:**
  Very strong — randomization reduces bias.

Level 3 – Cohort Study

- **Definition:**
  Researchers follow a group of people (a "cohort") over time to see who develops a disease.
- **Example:**
  Following 1,000 smokers and 1,000 non-smokers for 10 years to study lung cancer risk.
- **Strength:**
  Good for finding causes, but cannot fully prove them.

Level 4 – Case-Control Study

- **Definition:**
  Compares people with a disease (cases) to those without it (controls) and looks backward to find possible causes.
- **Example:**
  People with lung cancer vs. those without — how many smoked before?
- **Strength:**
  Useful when disease is rare; but memory bias may affect results.

Level 5 – Case Report / Case Series

- **Definition:**
  Detailed description of one or several patients with an unusual condition or response.
- **Example:**
  A report of a patient who recovered from COVID-19 after an experimental drug.
- **Strength:**
  Helpful for discovering new ideas — but not strong evidence.

Level 6 – Expert Opinion

- **Definition:**
  Advice or interpretation by experienced doctors without research data.
- **Example:**
  A senior doctor believes a vitamin helps flu recovery.
- **Strength:**
  Useful when no studies exist — but weakest evidence; may include bias.

Comparing Strengths of Evidence

Level	Type of Evidence	Strength
1	Systematic review / meta-analysis	⭐⭐⭐⭐⭐
2	Randomized controlled trial	⭐⭐⭐⭐
3	Cohort study	⭐⭐⭐
4	Case-control study	⭐⭐
5	Case report / series	⭐
6	Expert opinion	⚪

What Makes Evidence Strong?

Strong evidence is:

✅ **Reliable:** gives same results if repeated.

✅ **Valid:** really measures what it says.

✅ **Fair:** free from personal or money bias.

✅ **Large and clear:** involves many patients and clear results.

What Is Evidence-Based Medicine (EBM)?

- **Definition:**
  EBM means using the **best current research evidence**, **your clinical skill**, and **the patient's values and needs** to make good medical decisions.

Formula:

- **EBM = Research + Clinical Experience + Patient Preference**

Why EBM Is Important

- Improves **patient safety and quality of care.**
- Reduces **errors** and **waste.**
- Keeps doctors updated with new science.
- Builds **trust** between doctor and patient.
- Without EBM, medicine becomes opinion, not science.

Example of EBM in Action

- Old habit: “Give antibiotics for every sore throat.” EBM: Evidence shows most sore throats are **viral**, not bacterial → no antibiotic needed.
- Using evidence avoids overuse and resistance.

The Steps of EBM

1. **Ask** a clear question (What is the best treatment?).
2. **Search** for reliable studies.
3. **Appraise** (judge) the evidence quality.
4. **Apply** results to your patient.
5. **Assess** the outcome and improve next time.

What Is Clinical Reasoning?

- **Clinical reasoning** means how a doctor thinks to solve a patient's problem.
  It includes collecting information, making sense of it, and deciding what to do.
- Example:
  Patient has chest pain → think of heart, lungs, muscles → test → decide diagnosis.

Steps in Clinical Reasoning

1. Gather data (history, exam, tests).
2. Think of possible causes (differential diagnoses).
3. Compare findings to support or reject each cause.
4. Choose the most likely diagnosis.
5. Plan and evaluate treatment.

Why Clinical Reasoning Needs Evidence

- Reasoning must be based on **facts**, not feelings. Evidence guides each step:
- Which test to order
- Which drug works best
- What risk is worth taking
- Critical thinking keeps the reasoning logical and fair.

Common Thinking Errors in Medicine

Doctors can make mistakes when:

- **Anchoring bias:** sticking to the first idea.
- **Confirmation bias:** seeing only what fits your belief.
- **Overconfidence:** thinking you can’t be wrong.
- **Availability bias:** judging by recent or dramatic cases.

Being aware helps you think more clearly.

Role of Critical Thinking

Critical thinking means:

- Asking “Why do I believe this?”
- Checking if the information is true and recent.
- Being open to change when new evidence appears.
- Using logic, not emotion.
- It is the **heart of scientific medicine.**

Summary

- **Evidence** = facts that support medical decisions.
- **Evidence has levels**: some strong, some weak.
- **Evidence-Based Medicine** joins science, experience, and patient values.
- **Clinical reasoning** uses evidence and logic to solve patient problems.
- **Critical thinking** helps doctors think safely and wisely.

Final Message

“Good doctors don’ t just memorize they ***question, reason, and use evidence.”***

Think critically.

Use evidence.

Care wisely. 🧠💙